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GEOSTATIONARY BENT-PIPE SATELLITE LINK BUDGET ANALYSIS & ATMOSPHERIC PROPAGATION STUDY

UZB451E – SPACECRAFT COMMUNICATIONS
TERM PROJECT REPORT

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Nomenclature and Abbreviations

Term	Definition
ACM	Adaptive Coding and Modulation
ASI	Adjacent-Satellite Interference
AWGN	Additive White Gaussian Noise
BER	Bit Error Rate
BCH	Bose–Chaudhuri–Hocquenghem error-correction code
BPF	Band-Pass Filter
BUC	Block Upconverter
C/N	Carrier-to-noise ratio
C/N_0	Carrier-to-noise-density ratio
C/I	Carrier-to-interference ratio
dB	Decibel
dBHz	Decibel-Hertz
dBW	Decibels referenced to one watt
DTH	Direct-to-Home satellite broadcasting
DVB-S2	Digital Video Broadcasting–Satellite–Second Generation
ECEF	Earth-Centered, Earth-Fixed coordinate frame
EIRP	Equivalent Isotropically Radiated Power
E_s/N_0	Symbol-energy-to-noise-density ratio
E_b/N_0	Bit-energy-to-noise-density ratio
FEC	Forward Error Correction
FSPL	Free-Space Path Loss
GEO	Geostationary Earth Orbit
GS1	Ground Station 1, Rome uplink station
GS2	Ground Station 2, Ankara downlink station
G/T	Antenna gain-to-system-noise-temperature ratio
HPA	High-Power Amplifier
HPBW	Half-Power Beamwidth
IF	Intermediate Frequency
IMD	Intermodulation Distortion
ITU-R	International Telecommunication Union Radiocommunication Sector
LDPC	Low-Density Parity-Check code
LNA	Low-Noise Amplifier
LNB	Low-Noise Block Downconverter
MODCOD	Modulation and Coding Scheme
OBO	Output Back-Off
$P_{1\text{dB}}$	One-decibel compression output-power point
QPSK	Quadrature Phase Shift Keying
RF	Radio Frequency

Term	Definition
SNR	Signal-to-Noise Ratio
SSPA	Solid-State Power Amplifier
T_{sys}	System noise temperature
TWTA	Traveling-Wave Tube Amplifier
ULPC	Uplink Power Control
VSAT	Very Small Aperture Terminal
XPD	Cross-Polarization Discrimination

1 Introduction

This report evaluates a one-way Ku-band geostationary satellite link from Rome to Ankara. The analysis begins with a conventional clear-sky link budget and then adds interference, atmospheric attenuation, receiver-noise variation, apparent GEO motion, and DVB-S2 adaptive coding and modulation.

1.1 System Architecture and Ku-Band Principles

A circular equatorial orbit at an altitude of approximately 35,786 km has a period close to one sidereal day. A controlled spacecraft in this orbit therefore remains near a fixed longitude as viewed from the ground, allowing fixed parabolic antennas to be used. The corresponding slant range is about 37,000–38,000 km for the sites considered here, which produces more than 200 dB of free-space path loss. Small station-keeping excursions are evaluated separately to determine whether they have a measurable effect on this link.

The payload is modeled as a transparent *bent-pipe* transponder. It filters, frequency-translates, amplifies, and retransmits the received uplink without demodulating the waveform on board.

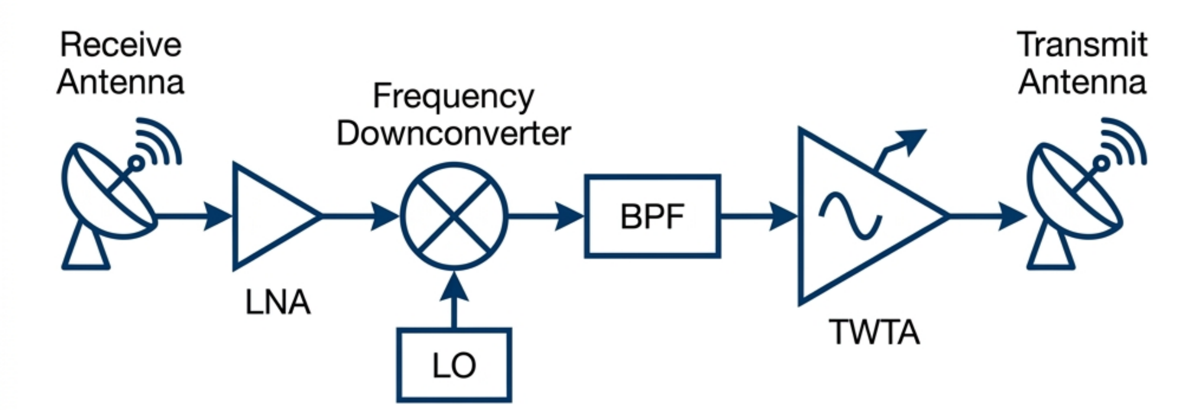


Figure 1: Functional block diagram of a classic transparent bent-pipe satellite transponder payload architecture.

Because the uplink waveform is not regenerated, uplink noise and interference are relayed with the wanted carrier. The uplink and downlink contributions must therefore be combined to obtain the end-to-end $(C/N_0)_{\text{total}}$.

The selected frequencies are 14.0 GHz for the uplink and 12.0 GHz for the downlink. A vertical linear-polarization case is used in the rain model, so ITU-R P.838-3 is evaluated with $\tau = 90°$; a horizontal assignment would require $\tau = 0°$. Ku-band permits useful antenna gain with relatively small apertures, but rain scattering and absorption are significant enough to require an explicit availability analysis.

1.2 Scenario Definition

The baseline is a one-way transparent link from Rome to Ankara through **Eutelsat HOTBIRD 13G**. Figure 2 defines the signal path used throughout the report.

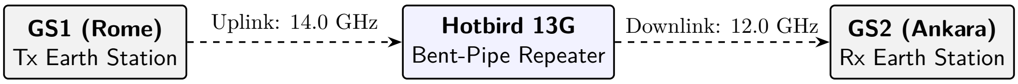


Figure 2: Transparent bent-pipe GEO link architecture between GS1 Rome, Hotbird 13G, and GS2 Ankara.

1.2.1 Satellite Selection Rationale

HOTBIRD 13G was selected for this Rome–Ankara broadcast case, not as a general ranking of GEO spacecraft. Its 13°E position gives useful elevation at both terminals, while Eutelsat publishes a Ku-band widebeam EIRP contour that supports a traceable report-level estimate [1]. Table 1 compares three named spacecraft using the same geometry routine; it is a preselection comparison rather than a second full link budget.

Table 1: Candidate satellite comparison for the Rome–Ankara bent-pipe scenario.

Candidate	Computed Geometry	Service and Evidence	Assessment
HOTBIRD 13G			
Eutelsat, 13°E	Rome: 41.60°, 37,659 km		
Ankara: 39.44°, 37,823 km	Ku-band DTH and video distribution across Europe and MENA, with an operator-issued 40–53 dBW widebeam map [1].	**Baseline.** The geometry is usable at both terminals, and the public footprint gives a traceable Ankara downlink EIRP estimate.	
ASTRA 1P			
SES, 19.2°E	Rome: 41.12°, 37,695 km		
Ankara: 41.68°, 37,652 km	Operational Ku-band widebeam spacecraft serving SES's European television neighbourhood; SES reports 80 transponders [2].	**Alternative.** The geometry is similar, but the consulted public source does not give an Ankara-specific contour for this link budget.	
Intelsat 39			
Intelsat, 62°E	Rome: 20.77°, 39,478 km		
Ankara: 34.92°, 38,187 km	C/Ku-band broadband, mobility, government, and video capacity across Europe, the Middle East, Africa, and Asia [3].	**Not used.** The lower Rome elevation gives a longer uplink path and a less favorable Ku-band fade case for this scenario.	

ASTRA 1P demonstrates that geometry alone does not uniquely determine the baseline: at 19.2°E it is also well balanced for Rome and Ankara. HOTBIRD 13G is retained because its operator brochure supplies a directly usable widebeam EIRP contour and its service context matches the modeled broadcast link. The SES fleet map is retained only as an orbital-slot cross-check for A1P at 19.2°E and IS-39 at 62°E, not as the technical source for payload or service claims [4]. Conversely, "Intelsat" denotes a fleet rather than one design point. Any Intelsat alternative would still require a named spacecraft, beam, polarization, EIRP contour, transponder plan, and receive (G/T) before a fair end-to-end comparison could be made.

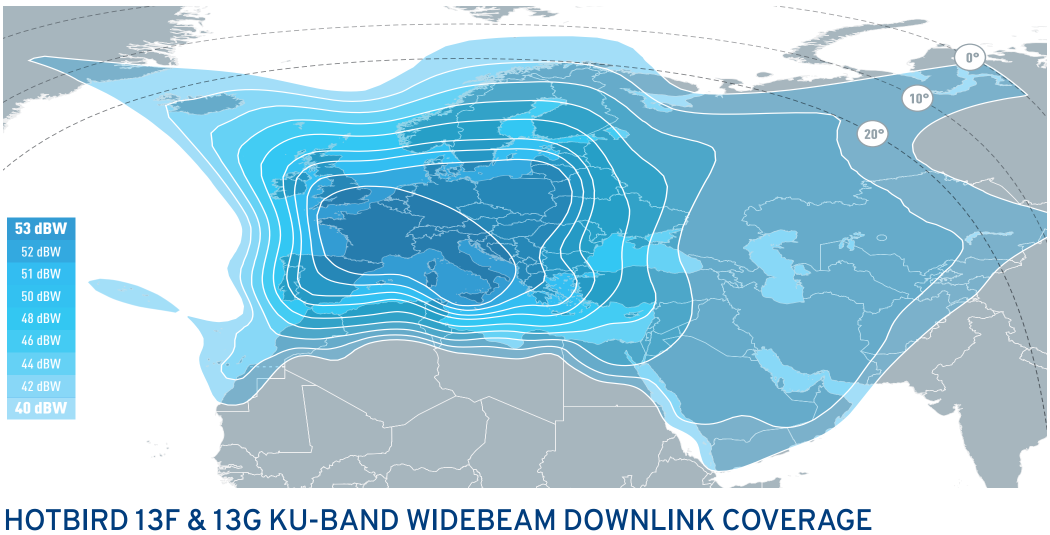


Figure 3: Official HOTBIRD 13F/13G Ku-band widebeam downlink EIRP contours. The map is reproduced from the Eutelsat HOTBIRD 13 East satellite brochure [1]. A conservative 46 dBW value is used for the Ankara downlink calculation rather than interpolating a higher value from the small-format contour map.

Figure 4: Author-created conceptual illustration of HOTBIRD 13G used to represent the transparent orbital repeater.

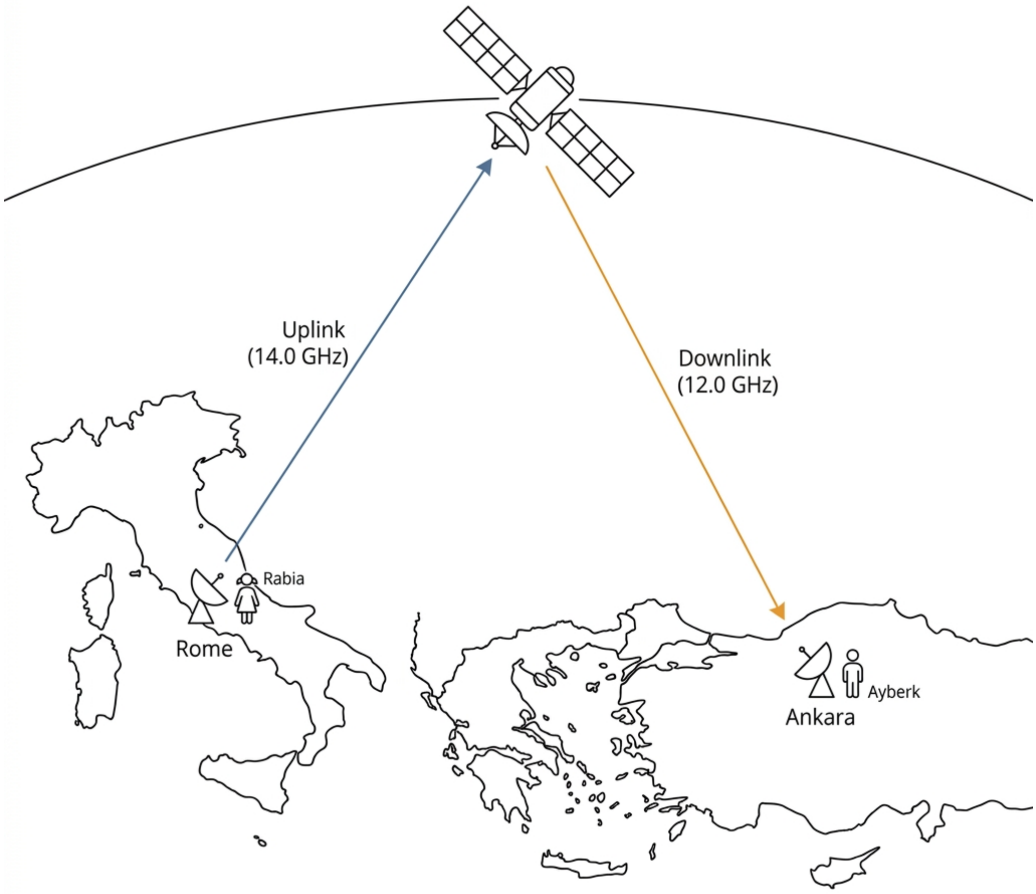


Figure 5: Architecture of the geostationary bent-pipe satellite link connecting GS1 Rome and GS2 Ankara via Eutelsat Hotbird 13G.

- **GS1 Rome:** The uplink site is at 41.9028° N, 12.4964° E. The antenna model uses a 2.4 m aperture, consistent with the Andrew Type 243 datasheet, which specifies 48.9 dBi gain at 14.3 GHz [5]. The uplink transmitter is modeled as a 20 W-class Ku-band BUC using the Norsat ATOMBKU020 $P_{1\text{dB}} \geq 43$ dBm rating as the rated boundary point [6]. Additional linear-output back-off is not credited separately in the uplink budget, and the model applies $L_{\text{tx,feeder}} = 1.0\,\text{dB}$.

- **HOTBIRD 13G:** The spacecraft is modeled at 13.0°E. Eutelsat's published wide-beam map contains 40–53 dBW contours; the Ankara downlink is assigned 46 dBW as a conservative contour-based estimate [1]. The selected transponder receive (G/T) is not published in the available operator material, so 3.0 dB/K remains an explicit engineering assumption.

- **GS2 Ankara:** The receive site is at 39.9334° N, 32.8597° E and 0.94 km altitude. The Triax TD88 sheet specifies a 95 cm by 85 cm offset reflector and 38.8 dBi gain at 11.7 GHz [7]. The calculation represents this aperture by an equivalent 0.9 m circular dish and obtains approximately 39.0 dBi at 12 GHz. The 150 K system noise temperature is an assumption because no LNB-specific noise-temperature measurement is available for the modeled terminal, and the receive feeder loss is modeled as $L_{\mathrm{rx,feeder}} = 0.5\,\mathrm{dB}$.

1.3 Computational Methodology

The project brief used MATLAB's Satellite Link Budget Analyzer as a reference workflow but allowed an equivalent implementation. The calculations were implemented in Python so that the assumptions, equations, test cases, CSV outputs, and plots could be kept in one reproducible codebase. The clear-sky equations follow Pratt's link-budget procedure [8]; the additional modules implement the stated ITU-R recommendation versions, a separate illustrative Monte Carlo weather model, interference estimates, dynamic receiver noise, apparent GEO motion, and DVB-S2 ACM.

Reproducible Analysis Package

The Python model, regression tests, generated CSV files, and plotting workflow are maintained in the public analysis repository:

`github.com/ayberkdt/link_budget_analysis`

1.4 Report Structure

Chapter 2 defines the geometry and clear-sky link equations. Chapter 3 describes interference, digital metrics, receiver-noise variation, and GEO motion. Chapter 4 gives the atmospheric and ACM models. Chapter 5 presents the numerical results, validation checks, sensitivity plots, and ablation study; Chapter 6 summarizes the design implications.

2 Link Design Equations

This chapter defines the geometry, antenna, propagation, and noise equations used in the clear-sky calculation. The link-budget sequence follows the procedure presented by Pratt, Bostian, and Allnutt [8]; the same equations are implemented in the Python model.

2.1 Slant-Path Geometry and Coordinate Transformations

Slant range, elevation, and azimuth are calculated by expressing the ground station and satellite positions in an Earth-Centered, Earth-Fixed (ECEF) frame.

From Kepler's third law, the orbital radius r_{sat} for a geosynchronous period T (one sidereal day, approximately 86,164 s) is:

$$r_{\text{sat}} = \sqrt[3]{\frac{\mu T^2}{4\pi^2}} \approx 42164.0\,\text{km} \tag{1}$$

where $\mu \approx 3.986 \times 10^5\,\text{km}^3/\text{s}^2$ is the Earth's standard gravitational parameter. Given the Earth's equatorial radius $R_e = 6378.137$ km, the satellite's altitude above the equator is approximately 35786 km.

For a ground station located at latitude ϕ, longitude λ, and altitude h above mean sea level, the ECEF position vector $\vec{R}_{\text{ground}}$ is:

$$\vec{R}_{\text{ground}} = \begin{bmatrix} (R_e + h)\cos\phi\cos\lambda \\ (R_e + h)\cos\phi\sin\lambda \\ (R_e + h)\sin\phi \end{bmatrix} \tag{2}$$

The geostationary satellite is modeled in the equatorial plane ($\phi_{\text{sat}} = 0°$) at longitude λ_{sat}, with Cartesian position vector:

$$\vec{R}_{\text{sat}} = \begin{bmatrix} r_{\text{sat}}\cos\lambda_{\text{sat}} \\ r_{\text{sat}}\sin\lambda_{\text{sat}} \\ 0 \end{bmatrix} \tag{3}$$

The line-of-sight vector from the ground station to the satellite is:

$$\vec{\rho} = \vec{R}_{\text{sat}} - \vec{R}_{\text{ground}} \tag{4}$$

The slant range d is the Euclidean norm of this vector:

$$d = \|\vec{\rho}\| = \sqrt{\rho_x^2 + \rho_y^2 + \rho_z^2} \tag{5}$$

The implementation uses a spherical Earth. A geodetic calculation based on the WGS-84 ellipsoid would instead use:

$$x = (N(\phi) + h)\cos\phi\cos\lambda \tag{6}$$

$$y = (N(\phi) + h)\cos\phi\sin\lambda \tag{7}$$

$$z = (N(\phi)(1 - e^2) + h)\sin\phi \tag{8}$$

where $e^2 = 2f - f^2$, $f \approx 1/298.257$, and $N(\phi) = R_e/\sqrt{1 - e^2\sin^2\phi}$. At GEO range, the spherical approximation changes the path length by much less than the uncertainty associated with the assumed RF parameters, so it is adequate for this project-level budget.

The elevation angle is obtained by projecting the unit line-of-sight vector $\vec{a}_\rho = \vec{\rho}/d$ onto the local upward unit vector:

$$\vec{u}_{\text{up}} = \begin{bmatrix} \cos\phi\cos\lambda \\ \cos\phi\sin\lambda \\ \sin\phi \end{bmatrix} \tag{9}$$

The elevation angle is therefore:

$$\theta = \arcsin\left(\vec{a}_\rho \cdot \vec{u}_{\text{up}}\right) = \arcsin\left(\frac{\vec{\rho} \cdot \vec{u}_{\text{up}}}{d}\right) \tag{10}$$

Azimuth is calculated from the local east and north components and is measured clockwise from true north:

$$Az = \arctan_2\left(\vec{a}_\rho \cdot \vec{u}_{\text{east}},\ \vec{a}_\rho \cdot \vec{u}_{\text{north}}\right) \tag{11}$$

2.2 Antenna Parameters and Aperture Efficiency

For a circular parabolic dish antenna of physical diameter D operating at carrier frequency f, the wavelength of the electromagnetic wave is $\lambda_w = c/f$, where c is the speed of light in vacuum. The peak boresight gain G in dBi is calculated as:

$$G = 10\log_{10}\left[\eta\left(\frac{\pi D}{\lambda_w}\right)^2\right] \qquad \text{[dBi]} \tag{12}$$

The aperture efficiency η collects the principal feed, illumination, blockage, and surface-error effects:

$$\eta = \eta_{sub} \cdot \eta_{spill} \cdot \eta_{taper} \cdot \eta_{surface} \tag{13}$$

where the factors represent blockage, spillover, illumination taper, and surface accuracy. A constant $\eta = 0.62$ is used when a frequency-specific datasheet gain is not supplied directly.

The Half-Power Beamwidth (θ_{HPBW}), which describes the angular width of the main beam and strongly affects adjacent-satellite interference susceptibility, is approximated by:

$$\theta_{\text{HPBW}} \approx 70^\circ \frac{\lambda_w}{D} \tag{14}$$

2.3 Free-Space Path Loss (FSPL) and Signal Spreading

Free-space path loss represents the reduction in power density caused by spherical spreading; it is not dissipative absorption by the vacuum.

Derived directly from the Friis Transmission Equation, the free-space path loss L_{FSPL} in decibel terms is expressed as:

$$L_{\text{FSPL}} = 20\log_{10}\left(\frac{4\pi d}{\lambda_w}\right) = 92.45 + 20\log_{10}(f_{\text{GHz}}) + 20\log_{10}(d_{\text{km}}) \qquad \text{[dB]} \tag{15}$$

For the present GEO geometry, FSPL is approximately 206–207 dB.

2.4 Additional System Losses: Pointing Error and Polarization Mismatch

The clear-sky budget also includes fixed allowances for pointing, polarization mismatch, feeder loss, and miscellaneous implementation effects.

Antenna Pointing Error Loss (L_{point}): Wind, installation tolerance, structural deformation, and orbital motion can offset the antenna boresight. For a small pointing error θ_{error}, a Gaussian main-beam approximation gives:

$$L_{\text{point}} \approx 12 \left(\frac{\theta_{\text{error}}}{\theta_{\text{HPBW}}} \right)^2 \qquad [\text{dB}] \tag{16}$$

Polarization Mismatch Loss (L_{pol}): A fixed 0.3 dB allowance is used for residual feed alignment and polarization mismatch. Vertical linear polarization is selected for the representative P.838 calculation, but an operational link would use the polarization assigned to the leased transponder. The Type 243 datasheet specifies 30 dB on-axis cross-polarization discrimination [5]; rain depolarization and imperfect alignment can reduce this isolation.

2.5 RF Link Power Budget and Receiver Figure of Merit

For each link segment, EIRP combines transmitter output power, antenna gain, and feeder loss:

$$\text{EIRP} = P_{\text{tx}} + G_{\text{tx}} - L_{\text{tx,feeder}} \qquad [\text{dBW}] \tag{17}$$

where P_{tx} is expressed in dBW, G_{tx} in dBi, and $L_{\text{tx,feeder}}$ in dB.

Receiver performance is represented by G/T:

$$\left(\frac{G}{T} \right) = G_{\text{rx}} - L_{\text{rx,feeder}} - 10 \log_{10} (T_{\text{sys}}) \qquad [\text{dB/K}] \tag{18}$$

where T_{sys} is the system noise temperature referred to the LNA input:

$$T_{\text{sys}} = T_{\text{ant}} + T_{\text{feeder}} + (F_{\text{LNA}} - 1)T_0 \tag{19}$$

Here T_{ant} includes sky and spillover contributions, T_{feeder} is the feeder contribution referred to the same plane, F_{LNA} is the LNA noise factor, and $T_0 = 290$ K. The static GS2 value of 150 K is a combined terminal assumption; it is not derived from a specified LNB noise figure.

The carrier-to-noise-density ratio at the receiver is:

$$\left(\frac{C}{N_0} \right) = \text{EIRP} + \left(\frac{G}{T} \right) - L_{\text{total}} - 10 \log_{10}(k) \qquad [\text{dBHz}] \tag{20}$$

where L_{total} encompasses FSPL, pointing misalignments, polarization losses, and atmospheric clear-sky absorption. The term $-10 \log_{10}(k) \approx 228.6$ dB represents Boltzmann's constant in decibels.

For a fixed information bit rate R_b:

$$\left(\frac{E_b}{N_0} \right) = \left(\frac{C}{N_0} \right) - 10 \log_{10}(R_b) \qquad [\text{dB}] \tag{21}$$

The fixed-rate margin is the difference between the available and required E_b/N_0:

$$M = \left(\frac{E_b}{N_0} \right) - \left(\frac{E_b}{N_0} \right)_{\text{req}} \qquad [\text{dB}] \tag{22}$$

2.6 Uplink-Downlink Coupling in Bent-Pipe Transponders

A transparent transponder does not regenerate the baseband signal. Uplink noise and interference therefore remain part of the retransmitted waveform.

The end-to-end $(C/N_0)_{\text{total}}$ is obtained by inverse-linear addition of the uplink and downlink terms:

$$\left(\frac{C}{N_0}\right)^{-1}_{\text{total}} = \left(\frac{C}{N_0}\right)^{-1}_{\text{uplink}} + \left(\frac{C}{N_0}\right)^{-1}_{\text{downlink}} \tag{23}$$

The combined C/N_0 is lower than either individual contribution. When one segment is much weaker than the other, the end-to-end result approaches the weaker segment and that path becomes the effective bottleneck.

3 Advanced Analysis Models

This chapter defines the optional impairment and diagnostic models added to the clear-sky calculation. Each model can be enabled independently, which also permits the ablation comparison reported in Chapter 5.

3.1 Adjacent-Satellite Interference (ASI) and IMD

Finite earth-station beamwidth permits energy from adjacent orbital slots to enter the receive chain and allows an uplink antenna to illuminate neighboring spacecraft. The resulting adjacent-satellite interference (ASI) depends on orbital spacing, antenna pattern, polarization, carrier loading, and relative EIRP. A multicarrier TWTA also generates intermodulation distortion (IMD) when operated close to saturation.

ASI and IMD are included as additional C/I terms so that the clear-sky result is not treated as purely thermal-noise limited.

The off-axis discrimination angle θ_{off} between the target satellite and the adjacent interferer is calculated using their respective ECEF position vectors. The off-axis gain $G_{\text{off}}(\theta_{\text{off}})$ in the direction of the interferer is estimated using a simplified internal off-axis antenna-envelope approximation:

$$G_{\text{off}}(\theta_{\text{off}}) = \min\left[G_{\max} - 3.0,\ 32.0 - 25\log_{10}(\theta_{\text{off}})\right] \tag{24}$$

This envelope is used for sensitivity analysis, not as an ITU-R regulatory mask or a coordination radiation pattern. The uplink, downlink, and IMD C/I terms are combined with the thermal C/N terms by inverse-linear addition:

$$\left(\frac{C}{N+I}\right)^{-1}_{\text{total}} = \left(\frac{C}{N}\right)^{-1}_{\text{uplink}} + \left(\frac{C}{N}\right)^{-1}_{\text{downlink}} + \left(\frac{C}{I_{\text{ASI,up}}}\right)^{-1} + \left(\frac{C}{I_{\text{ASI,down}}}\right)^{-1} + \left(\frac{C}{I_{\text{IMD}}}\right)^{-1} \tag{25}$$

In this project, the ASI/IMD analysis is used as a parametric impairment estimate rather than a regulatory coordination calculation. The adjacent satellites are modeled at 11.0°E and 15.0°E, corresponding to a nominal 2.0° orbital separation from the target 13.0°E slot. The adjacent-carrier activity factor is set to −3.0 dB, the antenna off-axis rejection follows the simplified envelope above, and the representative transponder IMD term is modeled as $C/I_{\text{IMD}} = 30$ dB. A formal coordination study would require operator-specific carrier plans, measured antenna radiation patterns, polarization reuse information, and transponder loading data.

Chapter 5 reports the resulting margin and separates ASI from the representative IMD term in the ablation table.

3.2 Hardware Impairments: Non-Linearity and Phase Noise

The transponder is not an ideal linear amplifier. TWTA compression, AM/PM conversion, and modem imperfections are represented only through simplified engineering terms:

- **Output Back-Off (OBO):** A deliberate reduction of the operating point below saturation to suppress non-linear distortion:

$$\mathrm{EIRP}_{\mathrm{active}} = \mathrm{EIRP}_{\mathrm{sat}} - \mathrm{OBO} \qquad [\mathrm{dBW}] \tag{26}$$

- **Implementation Margin (M_{impl}):** In this study, modem implementation effects are not included as a static RF path loss. Instead, a 1.0 dB DVB-S2 implementation margin is applied only during MODCOD threshold selection:

$$\left(\frac{E_s}{N_0}\right)_{\mathrm{usable}} = \left(\frac{E_s}{N_0}\right)_{\mathrm{available}} - M_{\mathrm{impl}} \tag{27}$$

An OBO value of 2.5 dB is discussed as a representative operating concept [8], but it is not subtracted separately from the published footprint EIRP: the 46 dBW downlink input is treated as the active contour value. Non-linear degradation in the numerical budget is represented by the assumed $C/I_{\mathrm{IMD}} = 30$ dB term. A 1.0 dB modem implementation margin is applied only during DVB-S2 MODCOD selection, and the static RF implementation-loss field is set to 0 dB to avoid double counting.

3.3 Digital Baseband Performance Metrics

For coherent QPSK over AWGN, the uncoded bit-error probability is:

$$P_{b,\mathrm{uncoded}} = \frac{1}{2}\,\mathrm{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right) \tag{28}$$

The report's generic digital-metrics layer applies a 3 dB equivalent coding-gain assumption for sensitivity calculations. The separate ACM calculation does not use this shortcut; it compares the available E_s/N_0 directly with the selected DVB-S2 threshold table. Shannon capacity is reported as an upper bound:

$$C = B \log_2(1 + \mathrm{SNR}) \qquad [\mathrm{bps}] \tag{29}$$

This coding framework is the foundation for the DVB-S2 MODCOD ladder used by the ACM selector in Chapter 4.

3.4 Dynamic System Noise Temperature

The static budget uses a fixed T_{sys}. The dynamic model instead varies the antenna-noise contribution with elevation and with absorptive atmospheric loss. A lossy rain path both attenuates the carrier and increases the apparent sky temperature seen by the receive antenna.

Separating these effects avoids treating rain solely as a carrier-power loss. Chapter 5 reports the carrier attenuation first and then adds the receiver-noise increase as a separate ablation step.

The dynamic, elevation- and fade-dependent system noise temperature T_{sys} is modeled at the LNA input as:

$$T_{\text{sys}}(\theta, A_{\text{rain}}) = T_{\text{receiver}} + T_{\text{spillover}} + T_{\text{sky}}(\theta) + T_{\text{rain}}(A_{\text{rain}}) \tag{30}$$

where $T_{\text{sky}}(\theta)$ is the clear-sky noise at elevation angle θ, and T_{rain} is the rain thermal emission noise calculated using the radiative transfer equation:

$$T_{\text{rain}} = T_{\text{atm}}\left(1 - 10^{-A_{\text{rain}}/10}\right) \tag{31}$$

where $T_{\text{atm}} = 275$ Kelvin represents the physical atmospheric temperature of the lossy rain medium.

For example, 10 dB of absorptive loss gives $T_{\text{rain}} \approx 247.5$ K for $T_{\text{atm}} = 275$ K. This is the rain-emission contribution only; the total T_{sys} also depends on the receiver, spillover, and clear-sky terms and is calculated from the full expression above.

3.5 Apparent GEO Orbit Motion

The apparent spacecraft position is allowed to vary within a simplified station-keeping box:

$$\Delta\lambda(t) = A_{\text{ew}} \sin\left(\frac{2\pi t}{T_{\text{sidereal}}} + \phi_{\text{ew}}\right) \tag{32}$$

$$\Delta\phi(t) = A_{\text{ns}} \sin\left(\frac{2\pi t}{T_{\text{sidereal}}} + \phi_{\text{ns}}\right) \tag{33}$$

$$\Delta r(t) = A_{\text{r}} \sin\left(\frac{2\pi t}{T_{\text{sidereal}}} + \phi_{\text{r}}\right) \tag{34}$$

These offsets update the slant-range vector $\vec{\rho}(t)$. The model is intended to test sensitivity to a controlled GEO station-keeping box; it does not establish a tracking requirement, which would depend on the measured antenna pattern, pointing tolerance, and operator practice.

4 Atmospheric Propagation and ACM

This chapter describes the atmospheric attenuation, illustrative outage, and DVB-S2 ACM calculations used after the clear-sky budget.

Model scope: the software is version-locked to the recommendation editions cited in this report, including P.618-13. It implements the published analytical procedures but uses representative meteorological inputs and a self-contained rain-height fallback rather than extracting every site value from ITU digital maps. The results are engineering sensitivity points, not an operational availability guarantee.

4.1 ITU-R Slant-Path Attenuation Suite

4.1.1 Slant-Path Loss Mechanisms

The Earth–space path crosses the troposphere, where gases, cloud liquid water, refractive-index fluctuations, and precipitation affect the received carrier. At Ku-band, rain is the largest time-varying attenuation term at the selected per-path $p = 0.1\%$ point.

4.1.2 Model Scope and Inputs

The advanced branch calculates an absolute atmospheric attenuation for each path. When this branch is active, it *replaces* the 0.5 dB static atmospheric allowance on that path; the two quantities are not added. Uplink and downlink attenuation are evaluated separately because their frequencies, elevations, and terminal apertures differ.

4.1.3 Mathematical Formulation

The total atmospheric attenuation $A_{\text{total}}(p)$ in dB exceeded for an annual time percentage p ($0.001\% \leq p \leq 5\%$) is calculated by combining gaseous absorption (A_{gaseous}), cloud attenuation (A_{cloud}), tropospheric scintillation ($A_{\text{scintillation}}$), and rain attenuation (A_{rain}) according to *ITU-R P.618-13* [9]:

$$A_{\text{total}}(p) = A_{\text{gaseous}} + \sqrt{\left[A_{\text{rain}}(p) + A_{\text{cloud}}\right]^2 + A^2_{\text{scintillation}}(p)} \tag{35}$$

The point rainfall rate $R_{0.01}$ is a representative P.837-based input [10]. The same $R_{0.01} = 42$ mm/h value is used at both terminals; a site-specific study would use separate mapped values for Rome and Ankara. P.839-4 defines rain height from a digital 0°C-isotherm map [11]; because that grid is not bundled with the project, the code uses a documented latitude approximation unless a mapped h_0 override is supplied. The frequency- and polarization-dependent k and α coefficients follow P.838-3 [12]. The specific attenuation is:

$$\gamma_{\text{R}} = k R_{0.01}^{\alpha} \tag{36}$$

The effective path length through the rain cell accounts for spatial localisation:

$$L_{\text{eff}} = L_{\text{s}}\, r_{0.01}\, v_{0.01} \tag{37}$$

where $r_{0.01}$ is the horizontal path reduction factor and $v_{0.01}$ is the vertical adjustment factor. The attenuation exceeded for 0.01% of the year is $A_{0.01} = \gamma_{\text{R}} L_{\text{eff}}$. For other time percentages p:

$$A_p = A_{0.01} \left(\frac{p}{0.01}\right)^{-[0.655 + 0.033 \ln p - 0.045 \ln A_{0.01} - \beta(1-p)\sin\theta]} \tag{38}$$

Gaseous attenuation uses dry air and water vapor specific zenith attenuations scaled by the slant-path elevation angle (*ITU-R P.676-12* [13]):

$$A_{\text{gaseous}} = \frac{\gamma_o h_o + \gamma_w h_w}{\sin\theta} \tag{39}$$

Cloud liquid water attenuation uses columnar liquid water M and specific attenuation K_l (*ITU-R P.840-8* [14]):

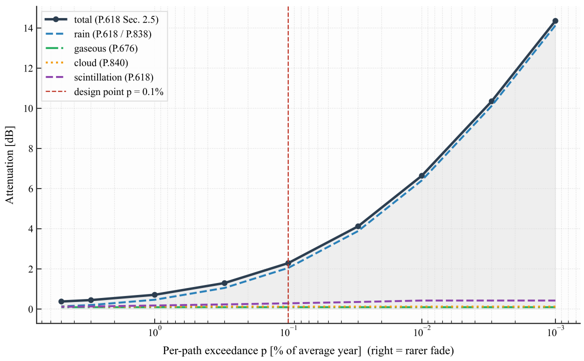
$$A_{\text{cloud}} = \frac{M K_l}{\sin\theta} \tag{40}$$

Scintillation uses the P.618-13 procedure with refractivity-related inputs consistent with P.453-14 [9, 15]:

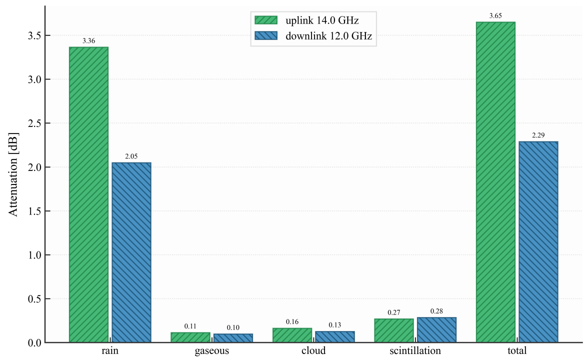
$$A_{\text{scintillation}}(p) = a(p)\sigma \tag{41}$$

4.1.4 Operational Impact and Results

At the per-path $p = 0.1\%$ point, the model gives 3.65 dB total uplink attenuation at 14 GHz and 2.29 dB at 12 GHz. Applying both fades simultaneously gives a conservative dual-site stress margin of 0.45 dB. This is not a statistically derived 99.9% end-to-end availability point: that claim would require a joint fade-occurrence or correlation model.



(a) Slant-path attenuation vs. per-path exceedance percentage.



(b) Attenuation breakdown at per-path $p = 0.1\%$.

Figure 6: ITU-R P.618 slant-path propagation impairments and specific contribution analysis for the Rome-Ankara link.

4.2 Monte-Carlo Rain Outage Simulation

4.2.1 Weather-State Model

The ITU-R design point gives attenuation exceeded for a specified percentage of an average year; it does not provide a chronological weather sequence. A separate Monte Carlo model is therefore used to generate an illustrative distribution of hourly margins.

4.2.2 Availability Estimation Rationale

This stochastic model is a sensitivity exercise based on assumed state probabilities and fade ranges. It is not fitted to measured precipitation records.

4.2.3 Mathematical Formulation

The simulation executes $N = 8760$ one-hour samples. At each t_i, Rome and Ankara draw separate weather states from the same illustrative probability mass function; a configuration switch can instead force correlated states for a deliberate double-hit stress experiment.

- S_{clear}**:** A small residual excess attenuation uniformly sampled between 0 and 0.15 dB.
- S_{light} **and** S_{moderate}**:** Small to medium fade depths drawn from uniform attenuation ranges.
- S_{heavy}**:** Rare high-fade states that represent severe Ku-band rain attenuation.

The discrete state variable is therefore:

$$S_i \in \{S_{\text{clear}}, S_{\text{light}}, S_{\text{moderate}}, S_{\text{heavy}}\}. \tag{42}$$

For each generated hourly weather state, the active link margin is recalculated as:

$$M_i = \left(\frac{E_b}{N_0}\right)_i - \left(\frac{E_b}{N_0}\right)_{\text{req}}. \tag{43}$$

The outage indicator for each sample is:

$$\mathcal{O}_i = \begin{cases} 1, & M_i < 0, \\ 0, & M_i \geq 0. \end{cases} \tag{44}$$

The corresponding link availability is then estimated as:

$$A_{\text{link}} = 100\left(1 - \frac{1}{N}\sum_{i=1}^{N} \mathcal{O}_i\right)\%. \tag{45}$$

4.2.4 Operational Impact and Results

The output is a distribution of link margin under the assumed states and random seed. Successive samples are independent, so the model does not reproduce storm duration or temporal correlation; its reported availability is illustrative.

Note: This Monte Carlo model is an illustrative stochastic availability simulation based on assumed weather-state probabilities (light rain: 2.0%, moderate rain: 0.6%, heavy rain: 0.1%), not a site-calibrated meteorological time-series model derived from measured annual precipitation data. Results should therefore be interpreted as *estimated availability under the assumed stochastic weather-state model*, not as a prediction of actual annual availability.

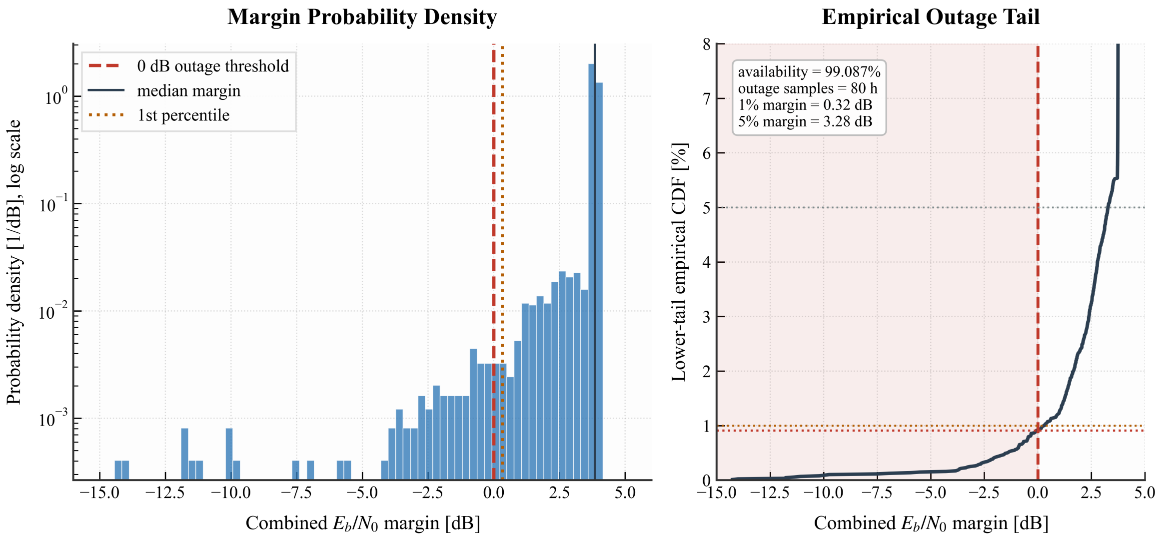


Figure 7: Monte Carlo annual margin distribution and outage-tail summary over 8,760 simulated hourly samples.

4.3 DVB-S2 ACM (Adaptive Coding & Modulation)

4.3.1 Adaptive-Throughput Context

A fixed MODCOD must be selected to meet its target availability with adequate margin. When channel conditions improve, that fixed operating point may leave capacity unused.

4.3.2 Fixed-Rate and ACM Metrics

The fixed-rate link margin and the ACM throughput analysis are treated as separate performance views. The former evaluates closure against a prescribed E_b/N_0 threshold, whereas the latter maps the available E_s/N_0 to DVB-S2 MODCOD thresholds and net throughput for a specified carrier bandwidth and roll-off factor. The two metrics should not be interpreted as the same quantity.

ACM maps the measured E_s/N_0 to a modulation and code rate. Higher-efficiency modes are selected when margin is available; more robust modes are selected during fades.

4.3.3 Mathematical Formulation

The receiver continuously monitors $(E_s/N_0)_{\text{available}}$ and dynamically selects the highest spectral efficiency η (in bits/symbol) from the DVB-S2 MODCOD ladder and its ideal QEF AWGN E_s/N_0 table [16] that satisfies:

$$\left(\frac{E_s}{N_0}\right)_{\text{available}} \geq \left(\frac{E_s}{N_0}\right)_{\text{MODCOD,req}} + M_{\text{impl}} \tag{46}$$

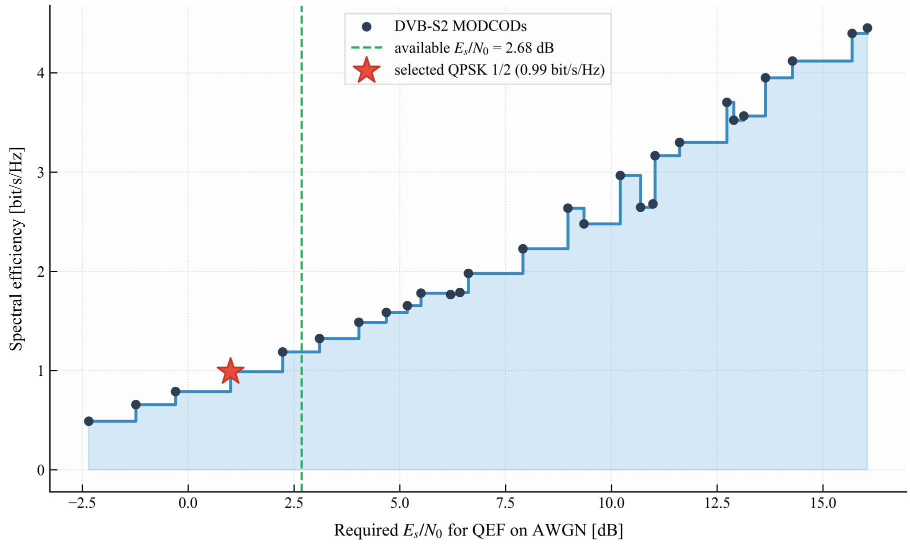


Figure 8: DVB-S2 MODCOD ladder: trade-off between spectral efficiency and required signal-to-noise ratio.

Using the usable symbol rate $R_s = B/(1 + \alpha_{\text{rolloff}})$, the active net throughput is:

$$R_{\text{net}} = \eta R_s \qquad \text{[Mbps]} \tag{47}$$

4.3.4 Operational Impact and Results

For a 36 MHz carrier with $\alpha = 0.20$, the symbol rate is 30 Msps. The implemented threshold table uses the ETSI normal-FECFRAME ideal values plus the 1.0 dB implementation margin in the selection rule. It selects QPSK 3/4 for the clear-sky thermal and interference cases and QPSK 1/2 for the coincident dual-site $p = 0.1\%$ stress point.

Table 2 lists the available and required E_s/N_0 values used by the selector.

Table 2: DVB-S2 ACM Consistency and Operating Point Verification

Case	**Available** E_s/N_0	**Selected MODCOD**	**Required** E_s/N_0	**Efficiency** η	**Net throughput**
Clear-Sky (Thermal)	$\approx$ 6.38 dB	QPSK 3/4	4.03 dB	1.487	$\approx$ 44.61 Mbps
With ASI & IMD	$\approx$ 5.92 dB	QPSK 3/4	4.03 dB	1.487	$\approx$ 44.61 Mbps
Dual-Site $p = 0.1\%$ **Stress**	$\approx$ 2.68 dB	QPSK 1/2	1.00 dB	0.989	$\approx$ 29.67 Mbps

Note: all points include a 1.0 dB implementation margin. Throughput is computed with $R_s = 30$ *Msps.*

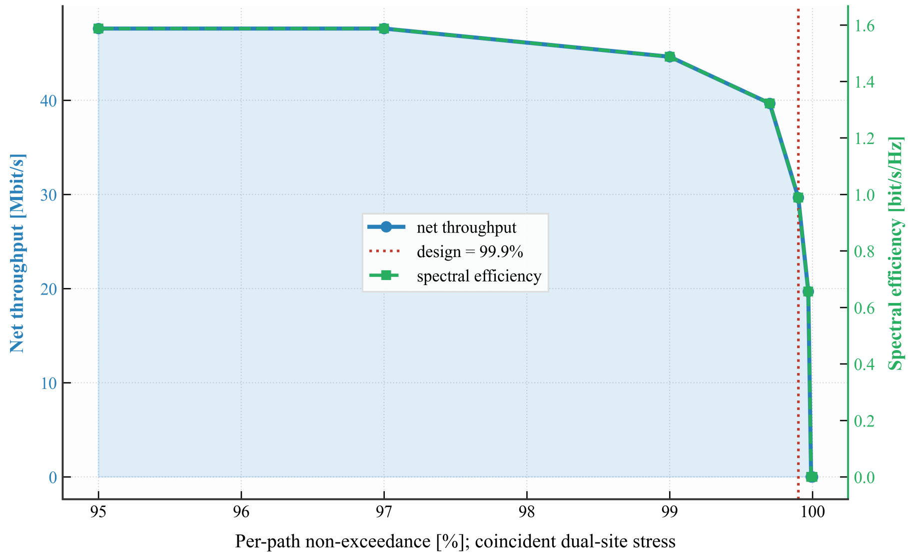


Figure 9: DVB-S2 ACM response over the coincident dual-site per-path exceedance sweep.

4.4 Uplink Power Control (ULPC)

Uplink power control is discussed as a mitigation option but is not implemented as a closed-loop simulation. An ideal first-order controller would increase earth-station transmit power by the estimated uplink fade:

$$P_{\mathrm{tx,dynamic}} = P_{\mathrm{tx,nominal}} + A_{\mathrm{rain,up}} \tag{48}$$

Here $P_{\mathrm{tx,dynamic}}$ and $P_{\mathrm{tx,nominal}}$ are in dBW, while $A_{\mathrm{rain,up}}$ is the uplink rain attenuation in dB. The command is subject to $P_{\mathrm{tx,dynamic}} \leq P_{\mathrm{tx,max}}$. Available HPA headroom, EIRP limits, ASI constraints, transponder loading, and control-loop delay limit the recoverable fade. No ULPC gain is credited in the reported margins.

5 Numerical Results and Discussion

This chapter reports the computed operating points, validation checks, sensitivity studies, and module-ablation results.

5.1 Static Clear-Sky Link Budget Results

The clear-sky calculation uses the Rome–HOTBIRD 13G–Ankara geometry defined in Chapter 1.

Table 3 distinguishes published hardware or operator values from model assumptions and explains how each value enters the calculation.

Table 3: Hardware and operator data traceability.

Item	**Published value**	**Model value**	**Treatment**
Andrew Type 243 antenna [5]	2.4 m aperture; 48.9 dBi at 14.3 GHz; 13.75–14.50 GHz Tx	2.4 m, $\eta = 0.62$	Computed gain is 48.86 dBi at 14.0 GHz, consistent with the datasheet.
Norsat ATOMBKU020 [6]	$P_{1\mathrm{dB}} \geq 43$ dBm; 14.0–14.5 GHz standard band	20 W-class rated boundary point	Model is based on the minimum $P_{1\mathrm{dB}}$ rating; no additional linear-output back-off is credited.
GS1 transmit feeder loss	Model assumption	$L_{\mathrm{tx,feeder}} = 1.0$ dB	Subtracted from the 20 W-class BUC output and antenna gain when computing uplink EIRP.
HOTBIRD 13F/13G widebeam [1]	13°E; official 40–53 dBW contour map	46 dBW at Ankara	Conservative map reading; not a transponder-specific guaranteed value.
Satellite receive (G/T)	Not published for the selected HOTBIRD transponder	3 dB/K	Explicit engineering assumption.
Triax TD88 [7]	95 cm × 85 cm; 38.8 dBi at 11.7 GHz	0.9 m equivalent circular aperture	$\sqrt{0.95 \times 0.85} = 0.899$ m; computed gain is approximately 39.0 dBi at 12 GHz.
GS2 receive feeder loss	Model assumption	$L_{\mathrm{rx,feeder}} = 0.5$ dB	Subtracted from receive antenna gain in the downlink (G/T) calculation.
GS2 receiver noise	LNB-specific combined T_{sys} unavailable	150 K static baseline	Explicit assumption; dynamic model is evaluated separately.
Fixed-rate threshold	Scenario design requirement	Required $E_b/N_0 = 7$ dB	Used only for the 10 Mbps fixed-rate closure test.

Table 4 gives the intermediate values from the software implementation of the Pratt-style calculation [8].

Table 4: Static Clear-Sky Link Budget Results (Rome - Ankara).

Parameter	**Uplink**	**Downlink**	**Unit**
Geometry			
Coordinates	41.90°N, 12.50°E	39.93°N, 32.86°E	–
Satellite longitude	13.00°E	13.00°E	–
Slant range (d)	$37,658.79$	$37,822.53$	km
Elevation angle (θ)	41.60	39.44	deg
Azimuth angle (Az)	179.25	209.37	deg
Central angle	41.91	43.85	deg
Transmitter			
Frequency (f)	14.00	12.00	GHz
Antenna diameter	2.40	–	m
Antenna gain	48.86	–	dBi
Transmitter power	13.01 (20.0 W)	–	dBW
Tx feeder loss ($L_{\text{tx,feeder}}$)	1.00	–	dB
EIRP	60.87	46.00 (Override)	dBW
Path losses			
Free-space path loss (FSPL)	206.89	205.59	dB
Pointing loss	0.50	0.50	dB
Polarization loss	0.30	0.30	dB
Clear-sky atmospheric loss	0.50	0.50	dB
Misc. loss	0.20	0.20	dB
Total link loss (L_{total})	208.39	207.09	dB
Receiver			
Antenna diameter	–	0.90	m
Antenna gain	–	39.00	dBi
Rx feeder loss ($L_{\text{rx,feeder}}$)	–	0.50	dB
Noise temperature (T_{sys})	–	150.00	K
Quality factor (G/T)	3.00 (Override)	16.74	dB/K
Performance			
Carrier-to-noise density (C/N_0)	84.08	84.25	dBHz
Carrier-to-noise ratio (C/N)	8.51	8.69	dB
Bit-energy ratio (E_b/N_0)	14.08	14.25	dB
Required E_b/N_0	7.00	7.00	dB
Link margin	7.08	7.25	dB

The end-to-end combined carrier-to-noise density ratio is computed using inverse-linear addition:

$$\left(\frac{C}{N_0}\right)_{\text{total,linear}} = \left(10^{-84.08/10} + 10^{-84.25/10}\right)^{-1} \tag{49}$$

$$\left(\frac{C}{N_0}\right)_{\text{total,dBHz}} = 10\log_{10}\left[\left(\frac{C}{N_0}\right)_{\text{total,linear}}\right] = 81.15\,\text{dBHz} \tag{50}$$

This gives $E_b/N_0 = 11.15$ dB and a 10 Mbps fixed-rate margin of **4.15 dB** against the

7 dB requirement.

Table 5 checks the implementation against an independent substitution of the same link-budget equations.

Table 5: Validation: Manual Pratt Calculation vs. Python Output.

Quantity	**Hand calculation**	**Python output**	**Difference**
Uplink FSPL	206.89 dB	206.89 dB	0.00 dB
Downlink FSPL	205.59 dB	205.59 dB	0.00 dB
Uplink EIRP	60.87 dBW	60.87 dBW	0.00 dB
Downlink G/T	16.74 dB/K	16.74 dB/K	0.00 dB
Uplink C/N_0	84.08 dBHz	84.08 dBHz	0.00 dB
Downlink C/N_0	84.25 dBHz	84.25 dBHz	0.00 dB
Uplink C/N	8.51 dB	8.51 dB	0.00 dB
Downlink C/N	8.69 dB	8.69 dB	0.00 dB
Combined C/N_0	81.15 dBHz	81.15 dBHz	0.00 dB
Combined E_b/N_0	11.15 dB	11.15 dB	0.00 dB
Combined margin	4.15 dB	4.15 dB	0.00 dB

5.1.1 Parametric Sensitivity and 2D Contour Map Analysis

Five contour plots show the principal static sensitivities and link-closure boundaries.

5.1.2 Downlink C/N_0 vs. GS2 Dish Diameter and Frequency

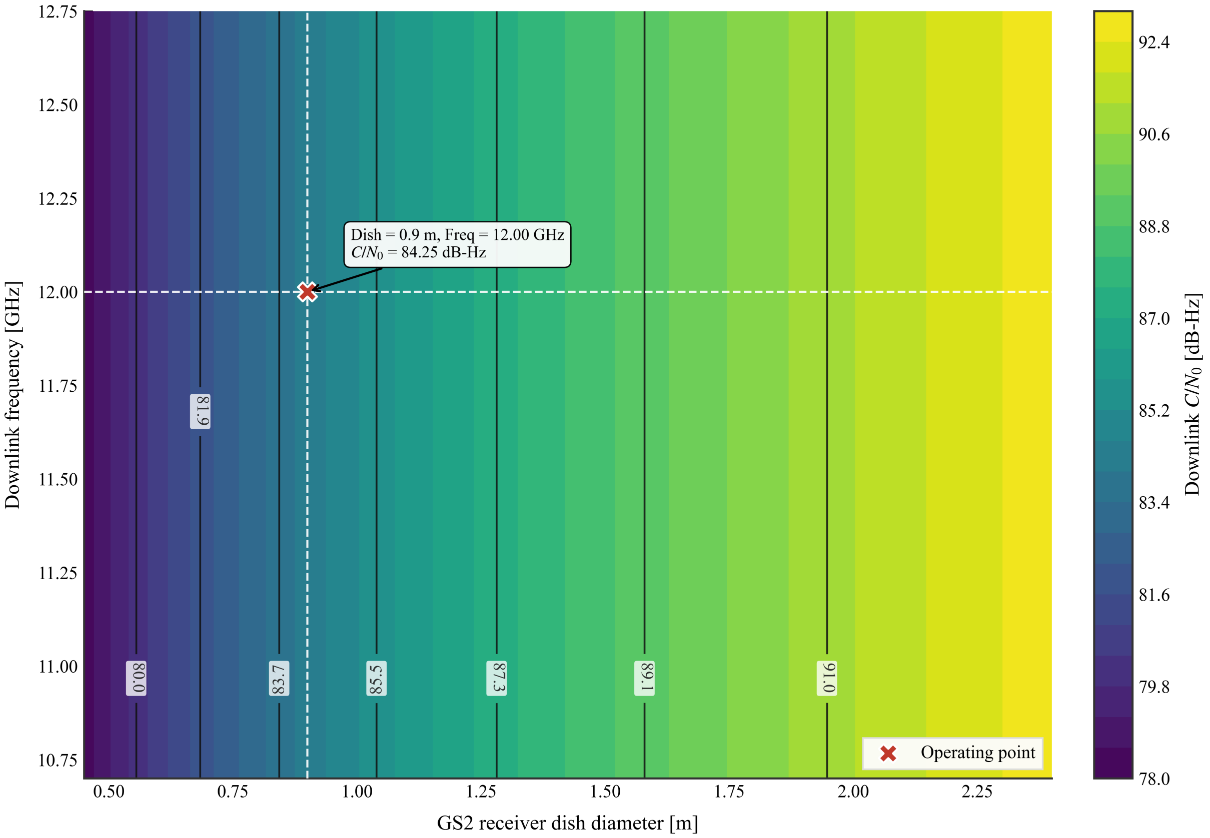


Figure 10: Downlink C/N_0 vs. GS2 dish diameter and frequency.

- **Purpose:** Evaluate downlink C/N_0 over receive-dish diameters of 0.5–2.4 m and frequencies of 10.7–12.75 GHz.
- **Result:** C/N_0 ranges from approximately 81.5 to 91.5 dBHz over the plotted domain. The nearly vertical contours show that aperture is the dominant variable.
- **Interpretation:** For fixed physical aperture and EIRP, receive gain and FSPL both contain an f^2 dependence, so their frequency terms largely cancel. Frequency becomes important when atmospheric loss, efficiency, or footprint EIRP also varies with band.
- **Design use:** The plot gives the minimum receive aperture needed for a specified C/N_0 under the fixed assumptions.

5.1.3 Downlink Margin vs. Satellite EIRP and GS2 T_{sys}

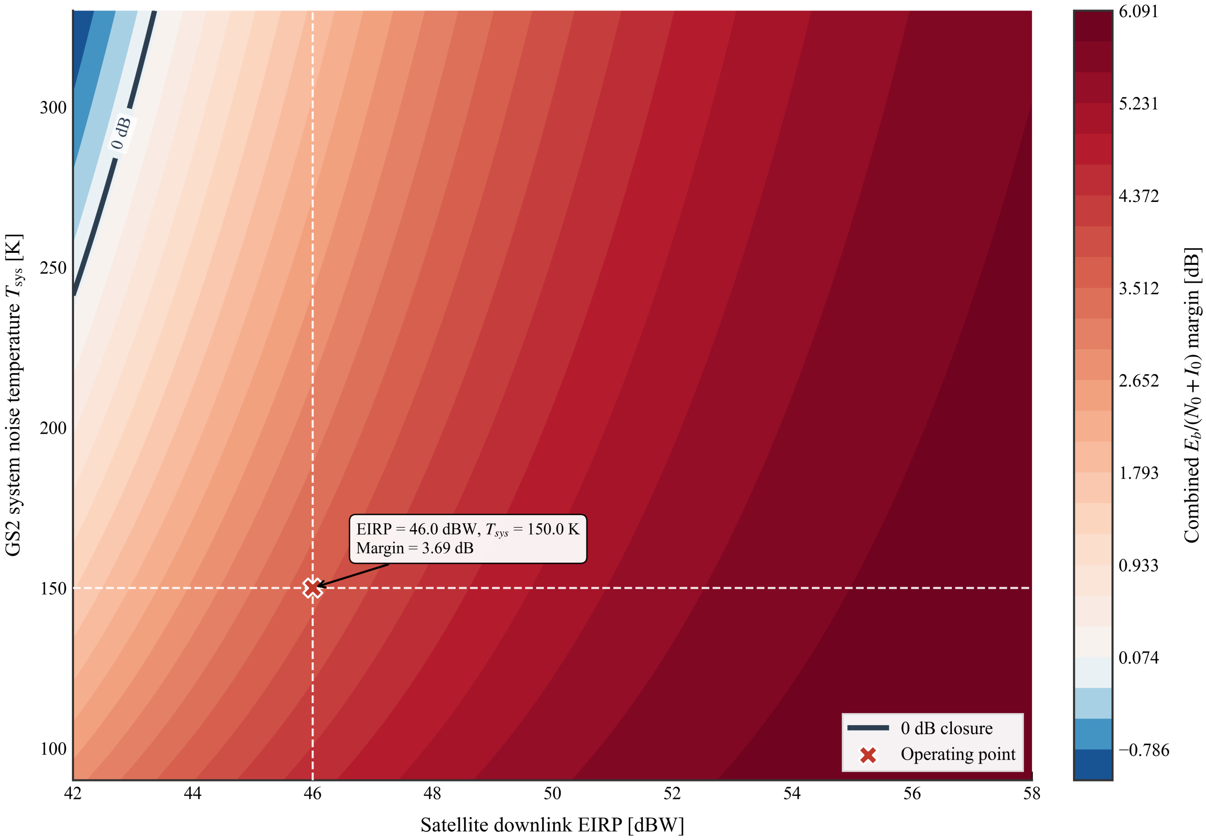


Figure 11: Combined effective margin vs. satellite EIRP and GS2 T_{sys}.

- **Purpose:** Map the 0 dB end-to-end margin boundary over 42–58 dBW downlink EIRP and 90–330 K receiver noise temperature, with ASI and IMD enabled.
- **Numerical result:** The 0 dB contour separates closed and open combinations. At $T_{sys} = 300$ K, closure requires approximately 45.5 dBW. The selected 46 dBW, 150 K point gives approximately 3.69 dB with ASI and IMD enabled.
- **Interpretation:** Margin increases one-for-one with EIRP in dB, while the receiver term varies as $-10\log_{10} T_{sys}$.
- **Design use:** The map quantifies the trade between leased downlink EIRP and receive-terminal (G/T).

5.1.4 Uplink C/N vs. HPA Power and GS1 Dish Diameter

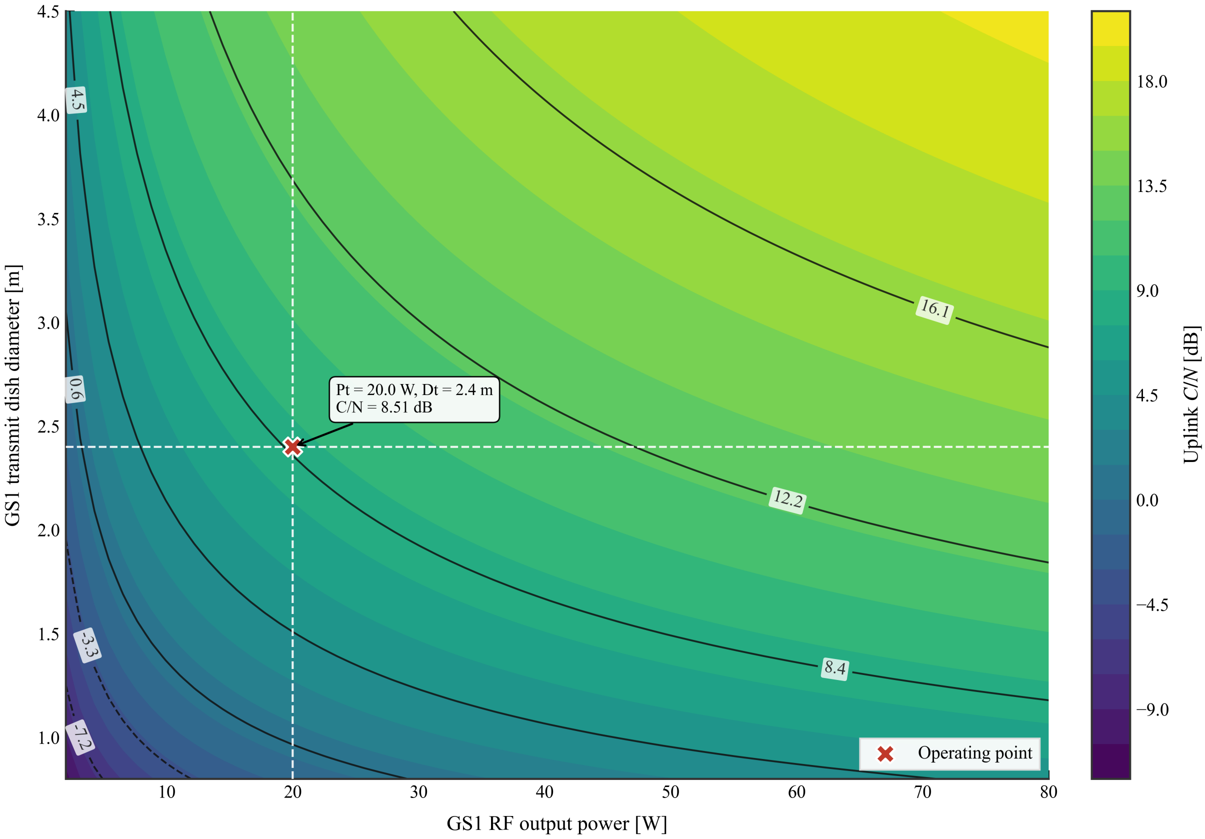


Figure 12: Uplink C/N vs. HPA power and GS1 dish diameter.

- **Purpose:** Compare 2–80 W HPA output with 0.8–4.5 m uplink aperture.
- **Result:** The 2.4 m, 20 W baseline gives $C/N \approx 8.51$ dB. A 12 dB target at the same aperture requires about 45 W. At 1.2 m, the additional 6 dB aperture penalty would require about 180 W, outside the plotted range.
- **Interpretation:** Doubling dish diameter adds approximately 6 dB of gain. For this case, aperture growth is more effective than increasing HPA output alone.
- **Design use:** The plot supports a cost and packaging trade between antenna size, amplifier rating, and off-axis discrimination.

5.1.5 Combined E_b/N_0 vs. Information Bit Rate and Bandwidth

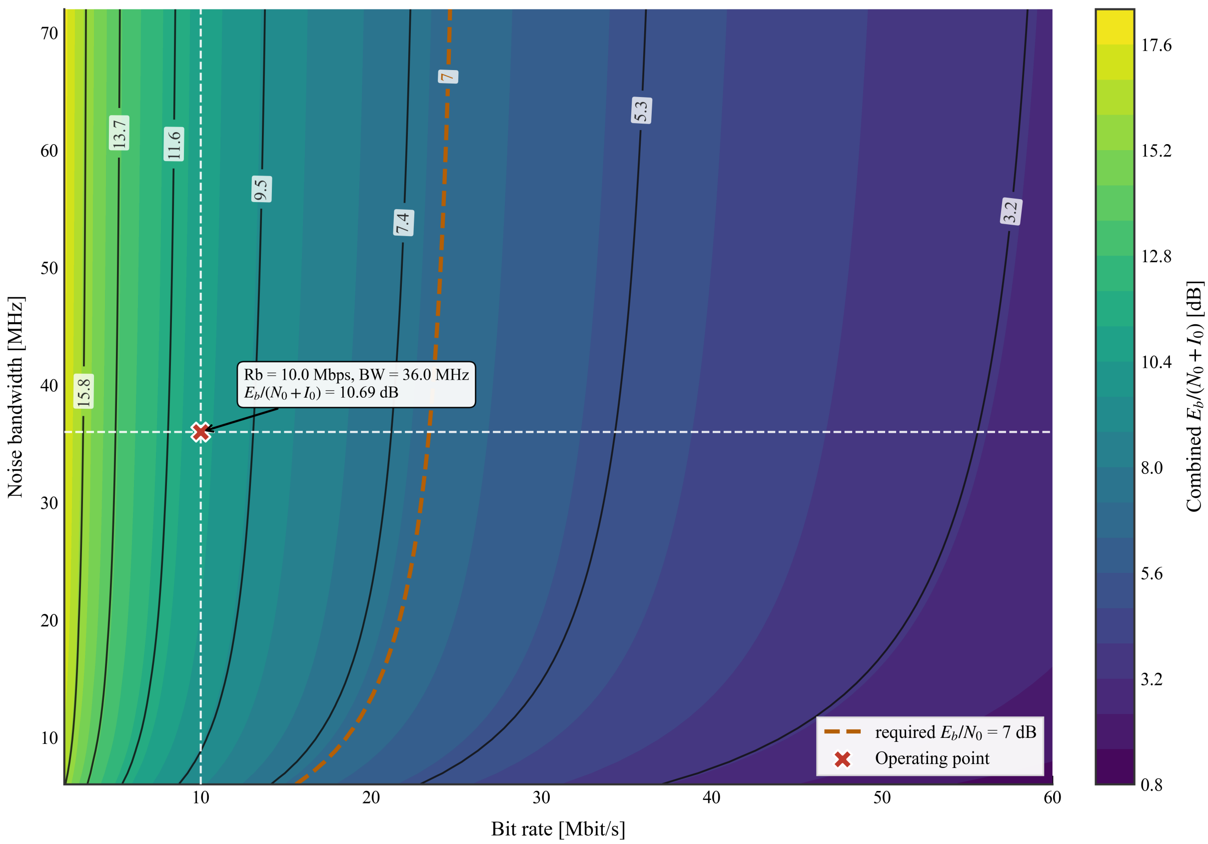


Figure 13: Effective end-to-end $E_b/(N_0 + I_0)$ vs. information bit rate and bandwidth.

- **Purpose:** Evaluate effective end-to-end E_b/N_0 over 2–60 Mbps and 6–72 MHz.
- **Result:** With $C/N_0 \approx 81.15$ dBHz, the noise-only E_b/N_0 falls from about 18.1 dB at 2 Mbps to 4.2 dB at 50 Mbps. At 10 Mbps it is 11.15 dB, or 10.69 dB with ASI and IMD.
- **Interpretation:** At fixed C/N_0, E_b/N_0 depends on bit rate, not directly on allocated bandwidth. Bandwidth instead limits symbol rate, occupied spectrum, and feasible MODCODs.
- **Design use:** The plot identifies the fixed-rate closure boundary for the stated 7 dB requirement.

5.1.6 GS2 Elevation Angle vs. Latitude and Longitude Separation

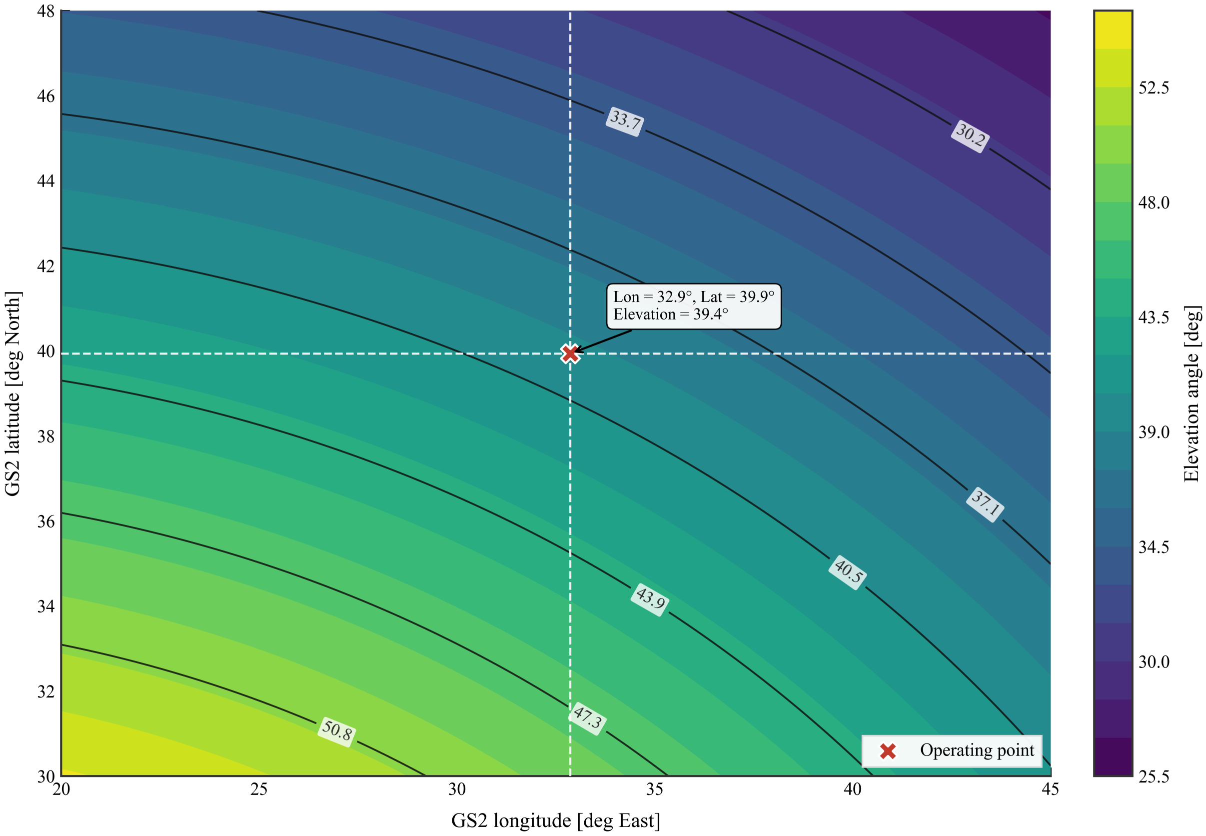


Figure 14: GS2 Elevation Angle vs Latitude and Longitude Separation.

- **Purpose:** Map GEO elevation angle over station latitude and longitude separation.
- **Result:** Elevation decreases as either latitude or longitude separation increases; 10-degree and 5-degree reference contours appear near the edge of the plotted domain.
- **Interpretation:** Lower elevation increases slant range and the atmospheric path length.
- **Design use:** The map indicates where additional atmospheric and pointing allowances become necessary.

5.2 Advanced Impairment and Dynamic Analysis

The advanced calculations retain the baseline hardware and geometry, changing only the active physical models.

5.2.1 Interference (ASI and IMD)

The modeled neighbors at 11°E and 15°E reduce the fixed-rate margin from 4.15 dB to 3.69 dB. Most of this change comes from ASI; the assumed 30 dB transponder IMD term contributes only 0.01 dB. These values are sensitivity assumptions, not coordination limits.

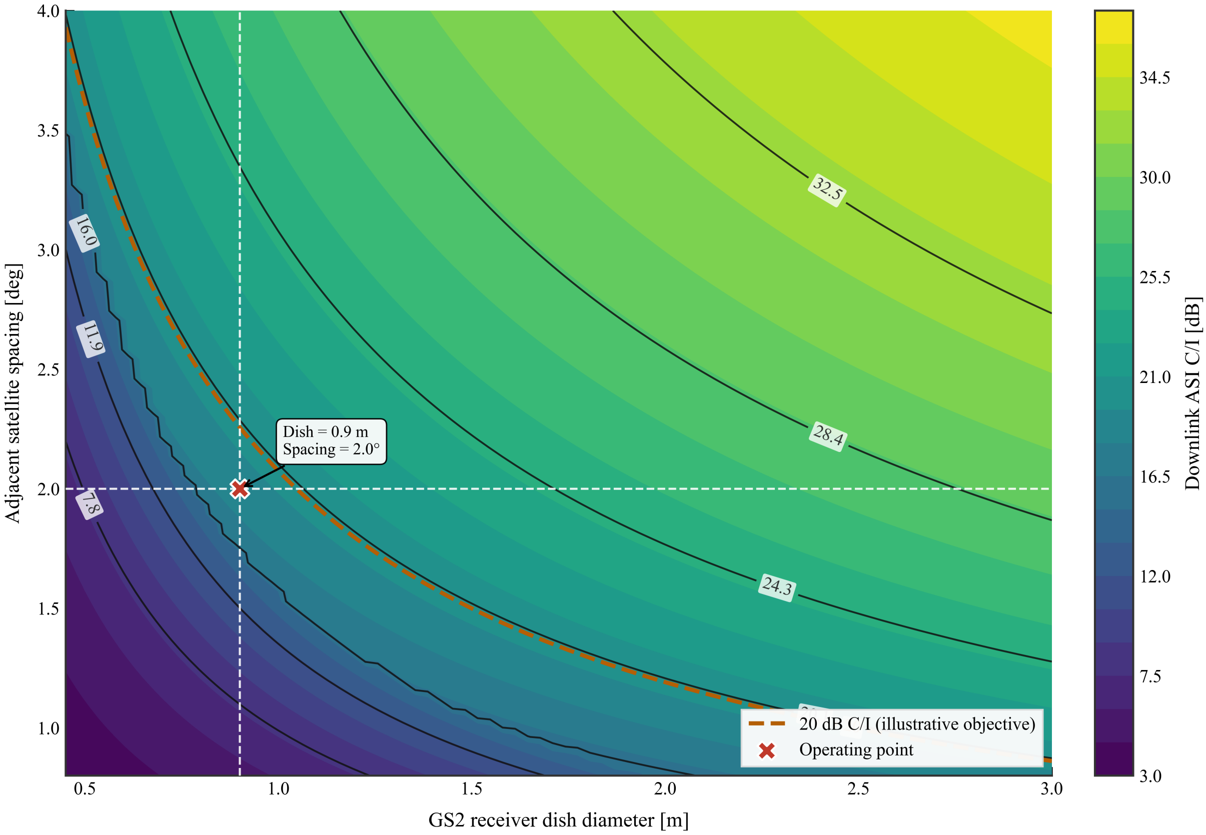


Figure 15: Downlink ASI C/I as a function of orbital spacing and receive-dish diameter.

5.2.2 Time-Varying Orbital Dynamics

The 48-hour station-keeping model produces a 3.98–4.00 dB impaired-margin envelope, only 0.017 dB peak-to-peak. The effect is negligible for this GEO budget and is presented as a model-consistency check.

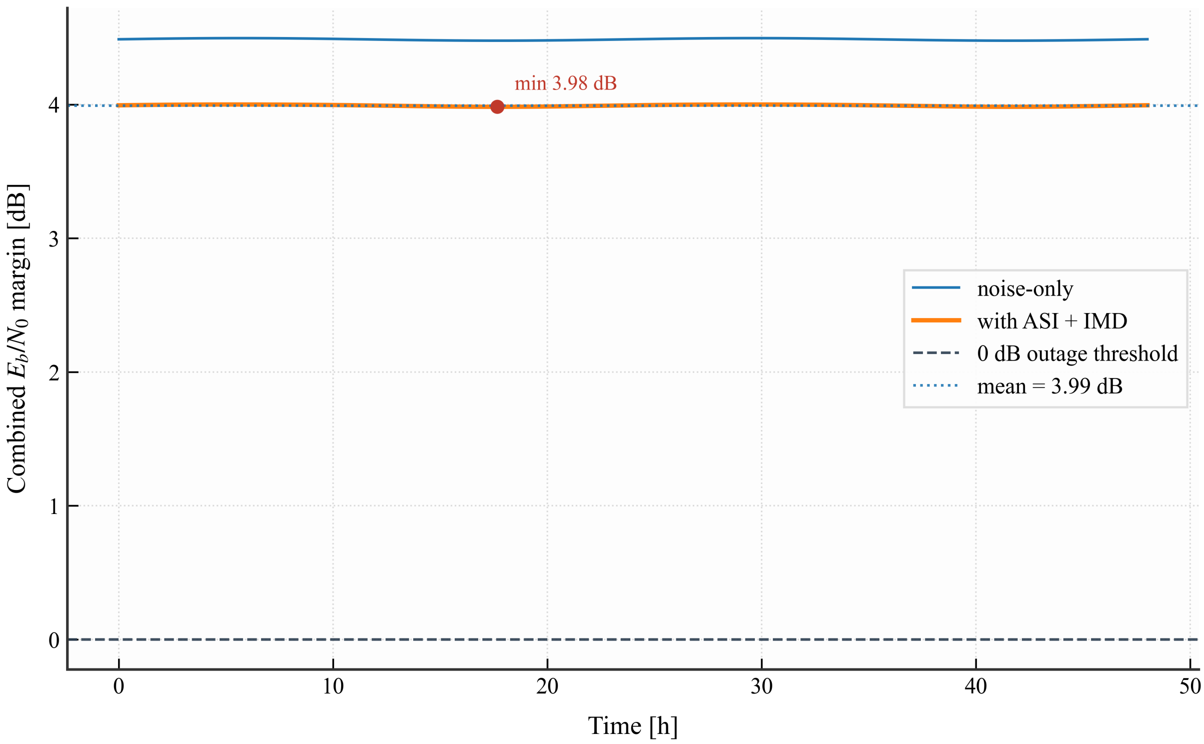


Figure 16: Combined link margin under the modeled apparent GEO motion. The solid curves show the noise-only and ASI/IMD-included margins over the 48-hour station-keeping cycle; the dashed line marks the 0 dB closure boundary and the dotted line marks the mean impaired margin.

5.2.3 Atmospheric p-Point Cases and ACM

At $p = 0.1\%$, the modeled totals are 3.65 dB on the uplink and 2.29 dB on the downlink. Table 6 separates the two single-path cases from the deliberately conservative coincident case. The latter is a stress point, not a 99.9% end-to-end availability estimate.

Table 6: Static and advanced operating-point comparison.

Case	UL/DL Loss	10 Mbps $E_b/(N_0 + I_0)$	Margin	ACM Mode	Interpretation
Clear sky, noise only	0/0 dB	11.15 dB	4.15 dB	QPSK 3/4, 44.6 Mbps	Static reference; interference excluded.
Clear sky with ASI and IMD	0/0 dB	10.69 dB	3.69 dB	QPSK 3/4, 44.6 Mbps	Impaired clear-sky snapshot.
Uplink-only $p = 0.1\%$	3.65/0 dB	9.17 dB	2.17 dB	QPSK 2/3, 39.67 Mbps	Rome fade applied; Ankara remains at its static allowance.
Downlink-only $p = 0.1\%$	0/2.29 dB	8.60 dB	1.60 dB	QPSK 3/5, 35.65 Mbps	Ankara fade and sky-noise emission applied.
Coincident dual-site $p = 0.1\%$	3.65/2.29 dB	7.45 dB	0.45 dB	QPSK 1/2, 29.67 Mbps	Conservative stress case; fixed-rate link still closes by 0.45 dB.

Reading the Two Performance Metrics

The 10 Mbps closure test uses $E_b/N_0 = C/N_0 - 10\log_{10} R_b$ and a 7 dB requirement. ACM instead uses $E_s/N_0 = C/N_0 - 10\log_{10} R_s$ with $R_s = B/(1+\alpha)$, then selects a MODCOD-specific threshold plus 1 dB implementation margin. ACM throughput is therefore a separate variable-rate service view; it does not add physical RF margin.

The independent-site Monte Carlo generator gives an illustrative annual closure fraction under assumed weather-state probabilities. Figure 17 shows the first 240 hours; it is not a measured or temporally correlated storm record.

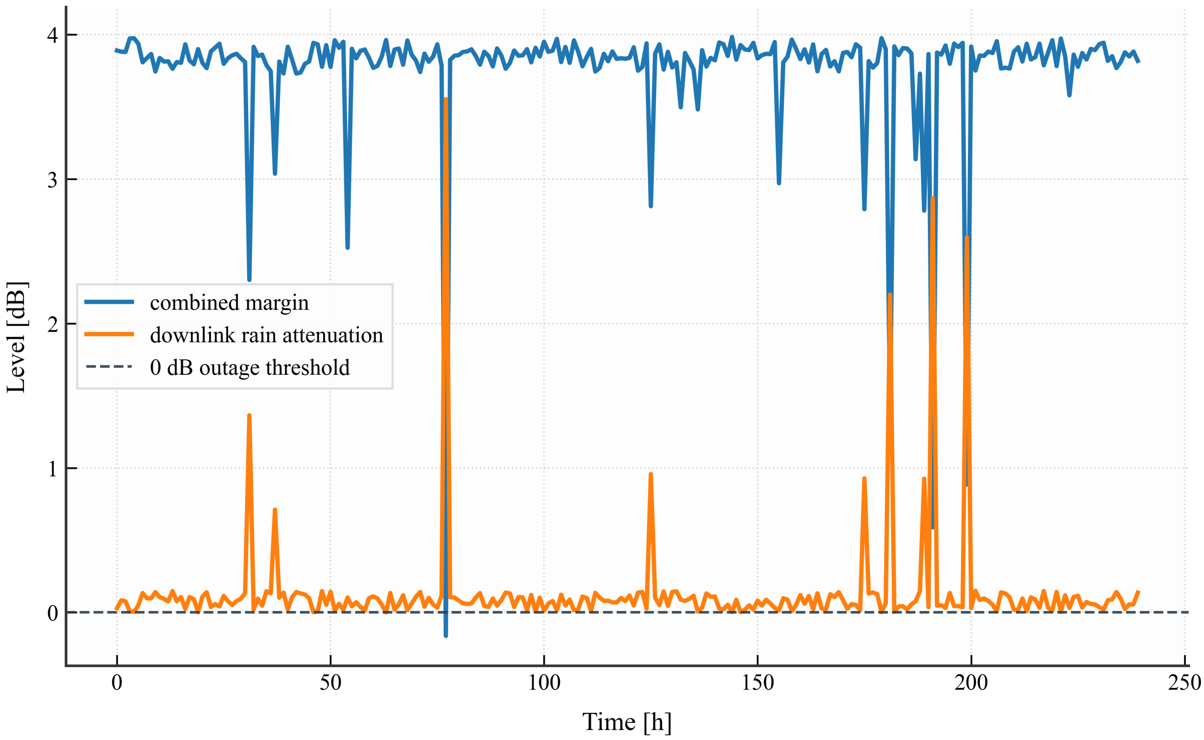


Figure 17: Illustrative margin and downlink attenuation samples with independent Rome and Ankara weather draws.

5.3 Module Ablation Study

The ablation chain in Tables 7 and 8 changes one modeling element at a time. Values are generated from the same functions exported to `outputs/ablation_summary.csv`.

Table 7: Controlled ablation of the advanced link-budget modules: numerical results.

Stage	Active Model Change	Margin	Increment	GS2 T_{sys}	UL/DL Atmospheric Total
A0	Thermal-noise-only static reference	4.15 dB	reference	150 K	0/0 dB
A1	Add adjacent-satellite interference	3.71 dB	−0.44 dB	150 K	0/0 dB
A2	Add the assumed transponder IMD term	3.69 dB	−0.01 dB	150 K	0/0 dB
A3	Replace each 0.5 dB static atmospheric allowance with gas, cloud, and scintillation	3.77 dB	+0.08 dB	150 K	0.43/0.41 dB
A4	Add rain carrier attenuation at per-path $p = 0.1\%$; keep T_{sys} fixed	1.36 dB	−2.41 dB	150 K	3.65/2.29 dB
A5	Replace fixed 150 K with the dynamic clear-sky receiver model; suppress atmospheric emission	1.62 dB	+0.26 dB	127 K	3.65/2.29 dB
A6	Add downlink atmospheric emission under the same carrier attenuation	0.45 dB	−1.17 dB	239 K	3.65/2.29 dB
A7	Apply the DVB-S2 ACM selector	0.45 dB	0.00 dB	239 K	3.65/2.29 dB
D1	Apply 48-hour station-keeping motion to the dynamic clear-sky case	3.98–4.00 dB	0.017 dB p–p	127 K	0/0 dB

Table 8: Controlled ablation of the advanced link-budget modules: engineering interpretation.

Stage	Engineering Finding
A0	Baseline fixed-rate link closes.
A1	ASI is the principal clear-sky interference term.
A2	IMD is minor for the selected 30 dB C/I.
A3	The modeled non-rain totals are slightly smaller than the static allowances.
A4	Rain-induced carrier loss is the dominant penalty.
A5	This isolates the receiver-temperature baseline change.
A6	The faded sky-noise term is substantial but no longer mixed with the baseline change.
A7	QPSK 1/2 gives 29.67 Mbps; physical margin is unchanged.
D1	Apparent GEO motion is negligible in this scenario.

The controlled sequence identifies three materially different effects: ASI costs about 0.44 dB, rain carrier attenuation costs 2.41 dB at the coincident stress point, and downlink atmospheric emission costs a further 1.17 dB. The non-rain replacement and dynamic clear-sky receiver model slightly improve the result relative to the original fixed allowances. This decomposition removes the earlier ambiguity in which a 150 K-to-dynamic-temperature change was incorrectly attributed entirely to rain emission.

6 Conclusion

This report evaluated a 14/12 GHz transparent GEO link from Rome to Ankara through HOTBIRD 13G using one reproducible Python model for the static budget, impairment cases, and ablation study.

The 10 Mbps clear-sky margin is 4.15 dB, falling to 3.69 dB with the selected ASI and IMD assumptions. At per-path $p = 0.1\%$, the uplink-only and downlink-only margins are 2.17 dB and 1.60 dB. Applying both fades simultaneously gives a conservative 0.45 dB stress margin. Because simultaneous distant-site fades were not assigned a joint probability, this value must not be presented as a 99.9% end-to-end availability guarantee.

The ablation study shows that rain carrier attenuation is the largest modeled penalty, followed by downlink atmospheric emission. ASI is smaller, IMD is minor for the stated assumption, and apparent GEO motion is negligible. The coincident stress case still closes the 7 dB fixed-rate requirement by 0.45 dB; the ACM selector independently chooses QPSK 1/2 at approximately 29.67 Mbps.

An operational design should replace the shared representative meteorological inputs with separate mapped or measured Rome and Ankara data and should use transponder-specific EIRP, (G/T), polarization, loading, and amplifier back-off. Those inputs are more important to the remaining uncertainty than the numerical precision of the present calculation.

ASTRA 1P has comparable Rome–Ankara geometry for a broadcast case, while Intelsat 39 serves a different coverage and service mix. HOTBIRD 13G remains the baseline because the public HOTBIRD material gives the clearest footprint evidence for the Ankara downlink estimate used in this report.

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